Electrodynamics
Independence Semester 2024

Homework 3

October 22, 2024

Problem 1

a) Show that the Bianchi identity $\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0$ can be written as

$$\partial_\mu \tilde{F}^{\mu\nu} = 0, \tag{1}$$

where $\tilde{F}^{\mu\nu}$ is the Hodge dual of $F_{\mu\nu}$, defined by $\tilde{F}^{\mu\nu} = \dfrac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$.

b) Show that the vector $V^\mu \equiv \tilde{F}^{\mu\nu} A_\nu$ has the property that

$$\partial_\mu V^\mu = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}. \tag{2}$$

c) Using $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\rho\sigma} = -2\delta^\mu_\alpha \delta^\nu_\beta + 2\delta^\nu_\alpha \delta^\mu_\beta$ show that the Hodge dual of $\tilde{F}_{\mu\nu}$ is equal to $-F_{\nu\mu}$.

Problem 2

a) Derive the equation of motion (Euler-Lagrange equation) that follows from the Lagrangian density

$$\mathcal{L} = -\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} - \frac{m^2}{8\pi} A^\mu A_\mu + J^\mu A_\mu, \tag{3}$$

where m is a constant, and $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$. This theory is a generalization of Maxwell's theory for vector (spin-1) bosons that possess mass, known as the *Proca theory*.

b) Is the Proca equation of motion invariant under gauge trnsformations $A'_\mu = A_\mu + \partial_\mu \lambda$?

c) Assuming that the 4-current J^μ is conserved (i.e. $\partial_\mu J^\mu = 0$), show that the Proca equation of motion implies that A^μ obeys $\partial_\mu A^\mu = 0$.

Problem 3

Show that the energy-momentum tensor of the electromagnetic field, $T_{\mu\nu} = \frac{1}{4\pi}\big(F_{\mu\rho}F^{\rho}_{\ \nu} - \frac{1}{4}F^{\rho\sigma}F_{\rho\sigma}\eta_{\mu\nu}\big)$, can be written as

$$T_{\mu\nu} = \frac{1}{8\pi}\big(F_{\mu\rho}F^{\rho}_{\ \nu} + \tilde{F}_{\mu\rho}\tilde{F}^{\rho}_{\ \nu}\big), \tag{4}$$

where $\tilde{F}^{\mu\nu}$ is the Hodge dual of $F_{\mu\nu}$, defined by $\tilde{F}^{\mu\nu} \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$.

Problem 4

Recall that the Hamiltonian for a particle of mass m and charge e in an EM field is given by

$$H = m\gamma + e\phi, \quad \text{where } \gamma = 1/\sqrt{1-v^2}. \tag{5}$$

a) If the particle moves in static EM fields, the Hamiltonian does not explicitly depends on time, i.e. $\partial H/\partial t = 0$. Then prove that the Hamiltonian is conserved, i.e., $dH/dt = 0$.

b) The time-independent quantity H is the energy $\mathcal{E}$ of the system: $\mathcal{E} \equiv H = m\gamma + e\phi$. Please recall from the notes that the relativistic Lorentz force equation can be written as $d\vec{p}/d\tau = e\gamma(\vec{E} + \vec{v}\times\vec{B})$. where $\vec{p} = m\gamma\vec{v}$. Using the above Lorentz equation to prove that

$$d\mathcal{E}_{\text{mech}}/dt = e\vec{v}\cdot\vec{E}, \tag{6}$$

where $\mathcal{E}_{\text{mech}} = m\gamma$.

c) Why does the magnetic field, described by a 3-vector potential $\vec{A}$, does not contribute to the conserved energy?

d) Let us consider the case where the particle is moving in static uniform $\vec{E}$ and $\vec{B}$ fields. Then show that the scalar potential and the 3-vector potential can be written as

$$\phi = -\vec{E}\cdot\vec{r}, \quad \text{and} \quad \vec{A} = \frac{1}{2}\vec{B}\times\vec{r}. \tag{7}$$

e) Suppose, for example, the uniform $\vec{B}$ field lies along the z-axis, i.e., $\vec{B} = (0, 0, B)$. Then find the 3-vector potential $\vec{A}$.

f) Now, $\vec{A}' = \vec{A} + \vec{\nabla}\lambda(\vec{r})$ will describe the same $\vec{B}$. Find the appropriate $\lambda(\vec{r})$ so that $\vec{A}'$ has the form $(A'_x, 0, 0)$. Also, find A'_x.